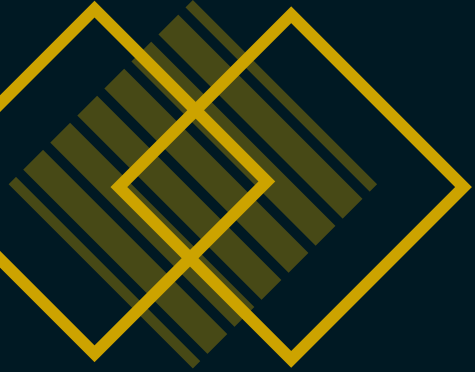




ROBO RASHTRA '26

चेक्राव्युह
गुडबोके



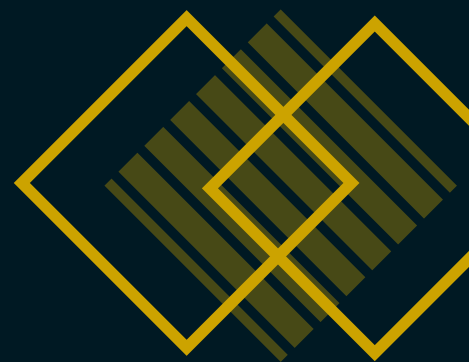
EVENT DESCRIPTION

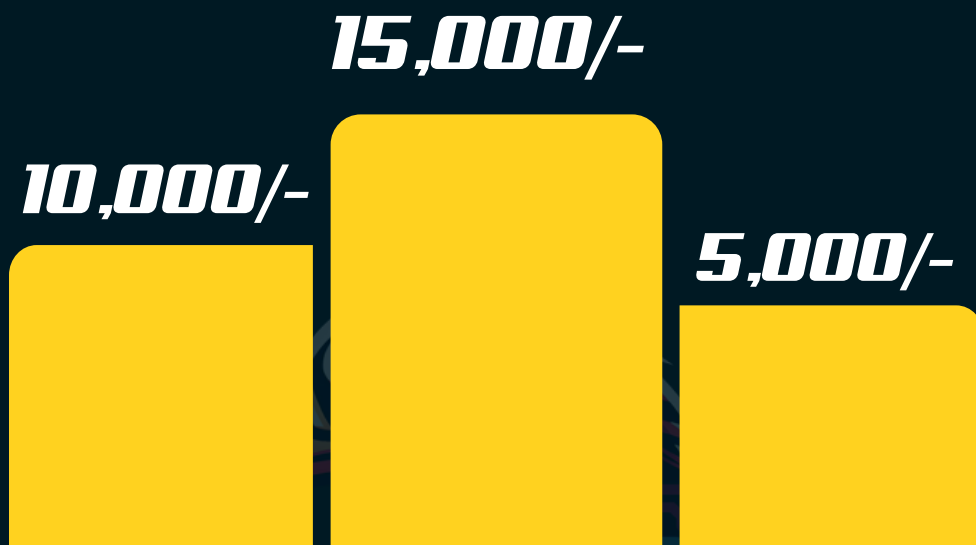
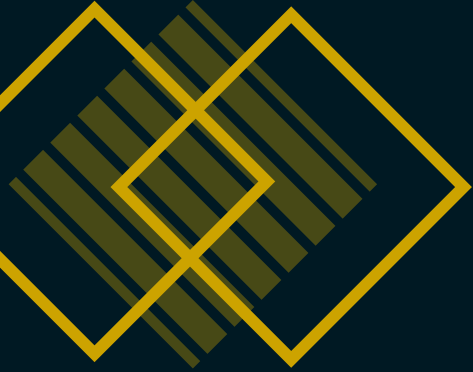
Chakravyuh is a robotics competition where students design an autonomous bot to complete two distinct stages:

Stage 1 – Line Following: The bot must follow a designated line track accurately.

Stage 2 – Maze Solving: After completing the line track, the bot must switch modes—either manually using a button or automatically—to solve a maze and find its way out in the minimum possible time.

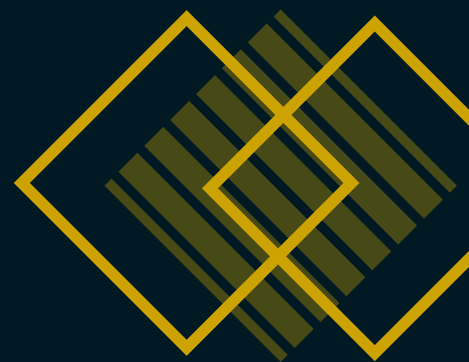
The winner will be decided based on the total completion time across both stages.

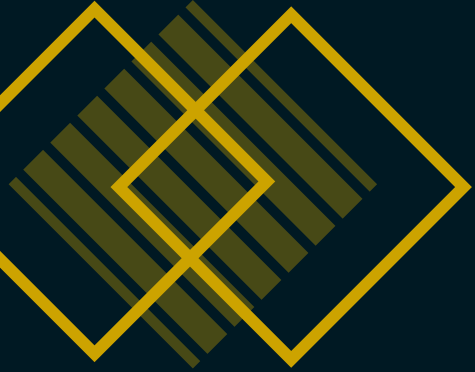




PRIZE POOL

30,000/-



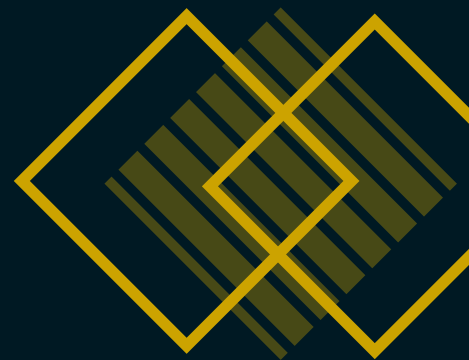


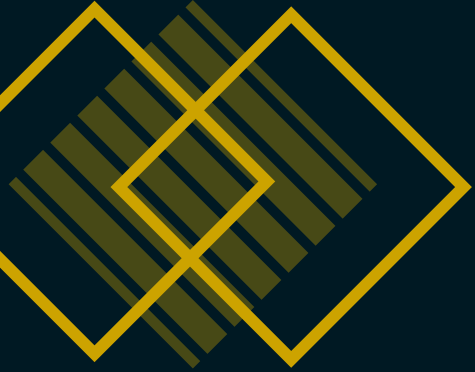
BOT SPECIFICATION

- **Size Restrictions:**
Length = 150 mm
Width = 150 mm
Height = 100 mm

A margin error of $\pm 5\%$ is acceptable.

- **Sensors:** Participants may use line sensors, IR sensors, or vision-based modules for navigation.
- **Control Mechanism:**
 - Bots must operate autonomously during the rounds.
 - The bot must not be connected to the internet or any external device using wireless communication
 - Manual intervention is allowed only to switch between modes (if not automated or else penalties will be added).

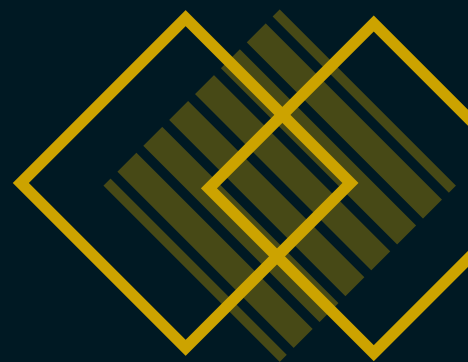
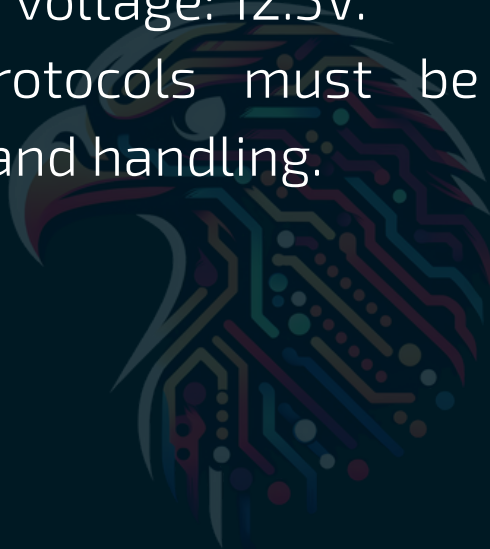


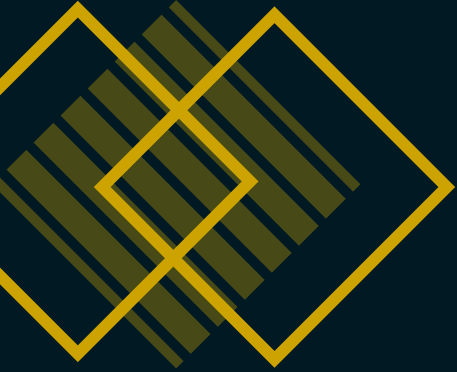


BOT SPECIFICATION

- **Power Supply:**

- Only onboard power sources allowed (rechargeable batteries or cells).
- Maximum voltage: 12.5V.
- Safety protocols must be followed during charging and handling.





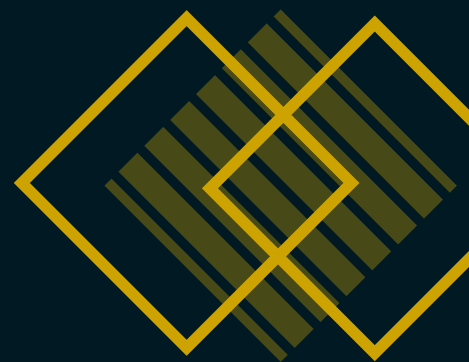
GAME PLAY

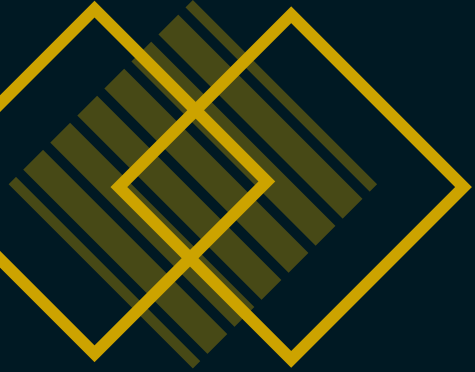
Starting Position:

- Bots will begin from the designated start zone in Stage 1.
- Judges will verify compliance before the round begins.

Stage 1 – Line Following:

- Bots must follow the given line track without deviation.
- If the bot deviates from the designated path, or gets stuck at a point it will be returned to the previous checkpoint(for which penalties will be added) and must continue from there.
- If the bot repeats the same loop three or more times, it will be returned to the previous checkpoint to continue its run.(for which penalties will be added)
- Time taken to complete this track will be recorded.





GAME PLAY

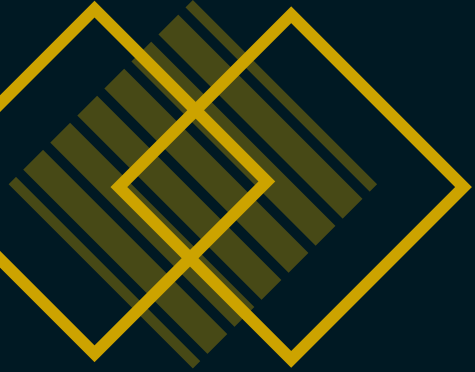
Stage 2 – Maze Solving:

- After Stage 1, bots must enter the maze.
- Switching to maze-solving mode can be manual (button press) or automatic.
- If the bot repeats the same loop three or more times, it will be returned to the previous checkpoint to continue its run.(for which penalties will be added)
- If the bot touches any walls during its run, penalties will be applied
- The objective is to reach the exit of the maze in the shortest possible time.

Time Limit:

- Each team will be allotted 10 minutes total for both stages combined.
- The bot must complete both stages within this time.

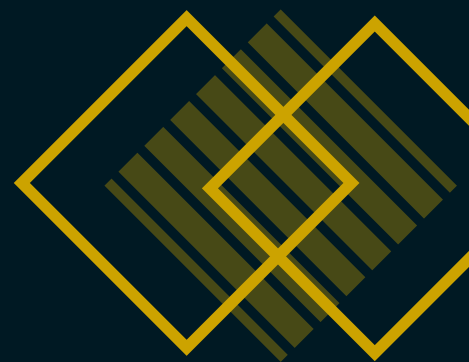


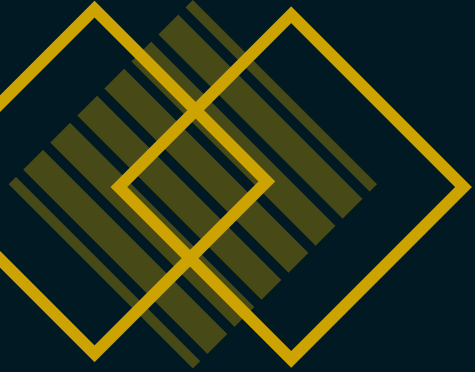


GAME PLAY

Emergency Handling:

- If the bot gets stuck, event officials may reposition it at the previous checkpoint.(for which penalties will be added)
- Time will continue to run; no pauses are allowed except technical timeout.
- In case of technical issues participants are allowed to make necessary adjustments in their bot,using the technical timeout.
- Technical timeout - 90 sec in case of technical issues with additional penalty.
- Each team can opt for only 2 timeouts during their entire gameplay.
- Once the game begins timer will not be stopped in any case unless the technical timeout is requested.





GAME PLAY

Rounds:

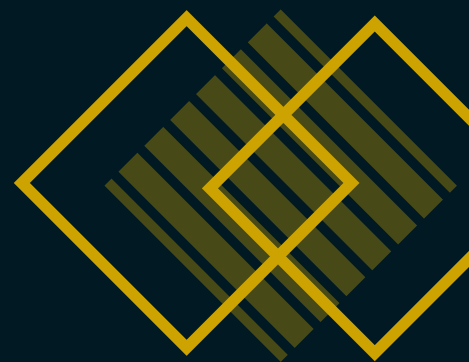
- Evaluation Round: Presentation and Compliance check of the bot specifications.
- Gameplay: Performance round (Line following + Maze solving).

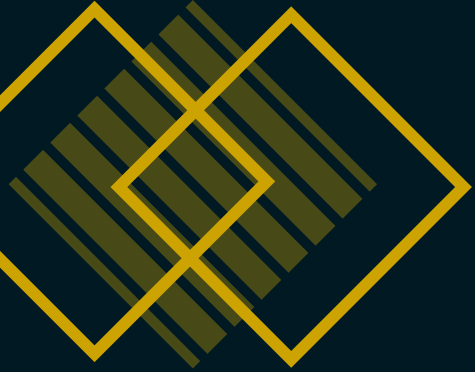
Winners:

- Winners will be declared based on total completion time.
- In case of a tie, accuracy in line following and maze-solving efficiency will be considered.

TEAM SPECIFICATIONS

Each team may consist of 2 - 4 members.





GAME RULES:

- Bots must adhere strictly to the technical specifications.
- No manual intervention is allowed during the run (except permitted mode switching or else penalties will be added).
- Damaging the arena or maze will lead to penalties.
- Judges' decisions are final and binding.
- Unsportsmanlike conduct or failure to comply with safety protocols will result in immediate disqualification.
- Participants may carry any additional or spare equipment and parts as required for their bot.
- Charging points will be provided at the venue for participants to recharge their bots or devices.

NOTE: Rules are subject to change. Any updates will be communicated before the event. Participants are advised to keep checking the website regularly.

